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Scientific Uniqueness & Literature Benchmarking

Benchmarking Against Reported Self-Cleaning Systems

System	Mechanism	Key Limitation	CASRE Differentiation
Superhydrophobic nano-rough coatings (silica/lotus type)	Micro/nano roughness, high contact angle	Fragile; dust entrapment; poor recoating	Smooth film; no nano-texture; chemistry-driven switching
Fluoropolymer coatings	Permanent low surface energy (–CF enrichment)	PFAS concerns; high cost; repaint issues	Fluorine-free; acrylic - compatible; regulatory aligned
TiO_2 photocatalytic coatings	UV-activated degradation, hydrophilicity	UV dependent; weak indoor performance	Non-photocatalytic; works interior & exterior
Silicone-modified paints	Static siloxane surface enrichment	Limited oil/household stain release	Dynamic dry–wet surface reconstruction

Core Scientific Innovation

Reversible Amphiphilic Surface Reconstruction

- Dry state: PDMS enrichment → reduced dust adhesion
- Wet state: Hydrophilic segment exposure → enhanced stain release
- Post-cleaning: Surface re-equilibrates to low-adhesion configuration

Mechanism is **chemistry-driven and reversible**

Advantages

- Fluorine-free and regulatory-aligned
- Smooth & recoatable
- Interior + exterior functionality
- Compatible with existing acrylic infrastructure

Defined End Use

Premium architectural coatings for high-traffic interiors and exposed urban facades requiring reduced maintenance cycles.

Literature Basis: Reported superhydrophobic silica systems, fluoropolymer coatings, TiO_2 photocatalytic coatings, and silicone-modified architectural paints.

Commercial Availability & Practical Differentiation

Current commercial categories

Existing Market Categories
- Standard acrylic emulsions
- Silicone-modified exterior
- Premium washable interior
- Fluoropolymer Specialty

Identified Gap
No single system integrates:
✓ Dry dust resistance
✓ Wet-triggered stain release
✓ Fluorine-free chemistry
✓ Smooth recoatable morphology

"CASRE integrates all four functionalities in a single adaptive coating platform."

CASRE Differentiation

✓ Broad-spectrum stain resistance
✓ Fluorine-free & regulatory aligned
✓ Recoatable smooth film
✓ Manufacturing compatible

Key Technical Risks
- Surface reconstruction rate may vary with temperature
- Requires graft-density optimization
- ~5–8% raw material premium

Maintains tintability, gloss options & ASTM-compliant durability.

Stain Resistance Scope – Interior & Exterior

Interior Stains – Easy-Clean Mode

Target Household Stains
•Tea, Coffee
•Turmeric, Mustard
•Oil / grease
•Ketchup / pickle
•Crayon / Pencil / marker / pen

Expected Performance
✓ Reduced surface adhesion (dry state)
✓ Limited stain penetration into film matrix
✓ ≥85–90% visible stain removal after damp wipe
✓ No permanent staining after controlled
✓ cleaning cycles

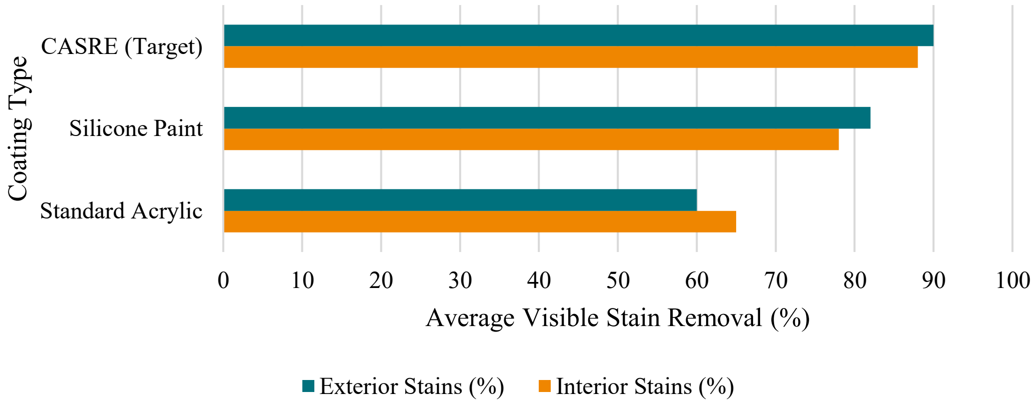
Cleaning Method
•Damp cloth
•Mild detergent (if required)

Exterior Stains – Self-Clean Mode

Target Environmental Contaminants
•Dust accumulation
•Mud splash
•Bird droppings
•Carbon soot / pollution
•Rust marks
•Betel leaf stains

Expected Performance
✓ Reduced dry dust adhesion
✓ Rain-triggered stain release
✓ No permanent staining after rain or
✓ low-pressure water wash
✓ Lower dirt pick-up index compared to
✓ standard acrylic benchmark

Interior vs Exterior Stain Removal Performance

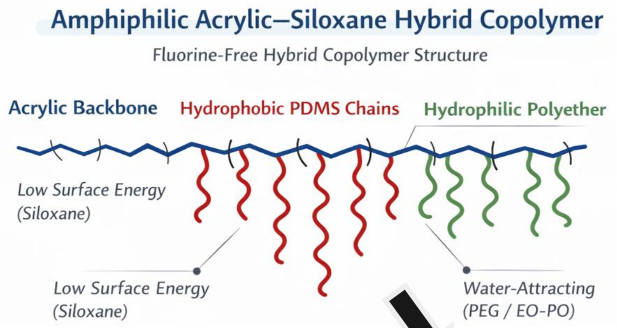


Processing Route & Formulation Strategy

Pre-Emulsion Preparation & Seeded Polymerization (70–80°C)

Siloxane Functional Monomer Grafting

PDMS-functional incorporation into acrylic backbone



Polymer Synthesis Stage

•Pre-emulsion of BA/MMA monomers
•APEO-free surfactant system
•Seeded emulsion polymerization (70–80°C)
•Particle size control

Pigment & Filler Dispersion

Functional Additive Incorporation

Final Processing, Quality Control & Packaging

•PVC range: 35–45%
•Rheology modifier addition
•Defoamer addition
•UV stabilizer incorporation
•Filtration & packaging

•Rutile TiO_2 (opacity & brightness)
•Surface-treated $CaCO_3$
•Fine treated silica
•High-speed disperser for uniform dispersion

•Amphiphilic acrylic–siloxane adaptive copolymer
•Silicone slip modifier (surface release)
•Barrier additive (stain penetration resistance)

Raw material safety, Storage & Application

Raw Material Safety & Regulatory Alignment
Safe Handling & Compliance

Binder System
- Water-based acrylic–siloxane hybrid
- Low VOC formulation
- Fluorine-free (no PFAS)

Additives
- Silicone slip modifier – low toxicity grade
- Barrier additive – non-reactive polymeric additive
- APEO-free surfactant system

Regulatory Alignment
- Aligned with architectural water-based coating standards
- No heavy metal pigments
- Compatible with green building certifications

Handling
- Non-flammable system
- Standard PPE (gloves, goggles) sufficient

Storage & Stability

Storage Conditions
- Store at 5–40°C
- Protect from freezing
- Avoid prolonged direct sunlight
- Shelf life target: 12–18 months
- Maintain sealed containers to prevent skin formation
- No special hazardous transport requirements

Compatible with existing paint warehouse infrastructure.

Application Technique

Surface Preparation
- Clean, dry, dust-free substrate
- Compatible with standard acrylic primer

Application Methods
- Roller / Brush / Airless spray
- Recommended DFT: 80–120 µm (2 coats)

Curing
- Ambient cure
- Touch dry: 30–60 min
- Recoat: 4–6 hrs

Cleaning
- Water-based clean-up

Development Challenges & Mitigation

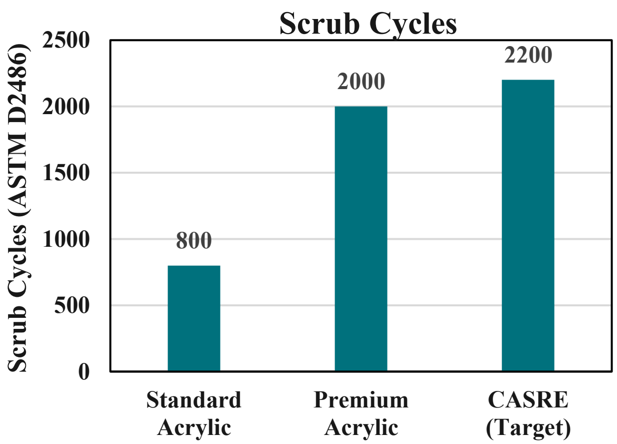
Challenge	Technical Risk	Mitigation Strategy
Surface Reconstruction Stability	Loss of switching efficiency over long-term exposure	Optimized graft density; controlled PDMS molecular weight; crosslinked acrylic stabilization
UV Degradation of Polymer Matrix	Chalking / performance decay in exterior exposure	HALS + UV absorber package; siloxane backbone improves UV resistance
Abrasion & Scrub Resistance	Surface smoothing may reduce mechanical durability	Balanced PVC (35–45%); optimized binder-to-pigment ratio; ASTM scrub validation
Climate Variability (Humidity / Temperature)	Switching kinetics may vary across climates	Accelerated climate cycling tests; formulation tuning for humidity response

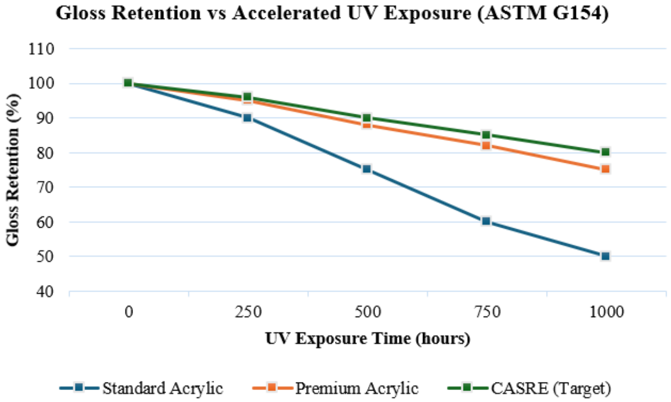
Interior (ASTM D2486)
- Scrub resistance: >2000 scrub cycles
- No ghost staining
- Repaint cycle: 5–7 years

Exterior (ASTM G154 / Accelerated UV)
- Anti-chalking
- Reduced dirt pick-up
- 5–7 years service life

"Designed to match or exceed premium acrylic durability standards while delivering adaptive self-cleaning functionality."

Expected Durability





Prototype Development & Validation Plan

Facilities Required

Polymer & Formulation Development
- Jacketed glass reactor (70–80°C emulsion polymerization)
- High-speed disperser
- Lab-scale mixing tanks
- Particle size analyzer

Performance Characterization
- Contact angle goniometer (dry/wet switching validation)
- ASTM D2486 scrub tester
- Gloss meter
- QUV accelerated weathering chamber
- Dirt pick-up resistance panel setup

Application & Field Testing
- Drawdown applicator
- Spray / roller setup
- Outdoor exposure racks (urban environment)

"All facilities aligned with existing architectural paint R&D infrastructure."

Development Timeline (12–18 Months)

Phase 1 (0–3 months) → Phase 2 (3–6 months) → Phase 3 (6–12 months) → Phase 4 (12–18 months)

Phase 1 (0–3 months)
- Polymer synthesis optimization
- Graft density tuning

Phase 2 (3–6 months)
- Coating formulation
- Stain resistance validation
- Initial scrub testing

Phase 3 (6–12 months)
- Accelerated UV & climate cycling
- Switching durability evaluation

Phase 4 (12–18 months)
- Pilot batch scale-up
- Field exposure
- Cost benchmarking

Proof of Concept – Validation Methodology

Validation Area	Test Method
Scrub Resistance	ASTM D2486 scrub test
UV Stability	QUV accelerated weathering (1000 hr cycle)
Stain Resistance	Standardized stain panel + wipe protocol
Dirt Pick-Up	Outdoor exposure & reflectance change (ΔL^*)

Scalability Strategy (Lab → Bulk)

Scale-Up Pathway

Lab Scale (1–5 L)
•Polymer synthesis validation
•Switching mechanism optimization
•Initial coating formulation

Pilot Scale (50–100 L)
•Batch reproducibility testing
•Graft density consistency
•Pigment dispersion uniformity
•Stability evaluation (shelf life)

Commercial Scale (1000+ L)
•Standard architectural paint plant integration
•Let-down stage additive incorporation
•No new reactor design required

Infrastructure Compatibility
•Water-based emulsion system
•No fluorochemicals
•No nano-texturing or exotic curing
•Compatible with standard high-speed dispersers
•Uses existing pigment/filler supply chain

Scale-Up Risks & Controls

Scale Risk	Control Strategy
Particle size variation	Controlled monomer feed rate
Graft uniformity issues	In-process QC monitoring
Batch viscosity variation	Standard rheology adjustment
Storage instability	Accelerated stability testing

"Fully compatible with conventional architectural paint manufacturing infrastructure."

Business Case & Market Feasibility

Target Market & Customer

Primary Segment (Entry Strategy)
•Premium interior easy-clean paints
•Urban exterior coatings (pollution-heavy cities)
•Hospitals, schools, airports, metro stations, Smart city infrastructure

Why Premium First?
•Customers already pay for washability
•Value perception is higher
•Easier margin absorption

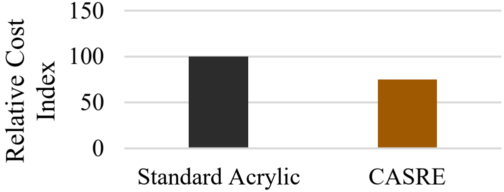
"Positioned within premium architectural coatings with adaptive easy-clean advantage."

Cost & Feasibility Cost Impact

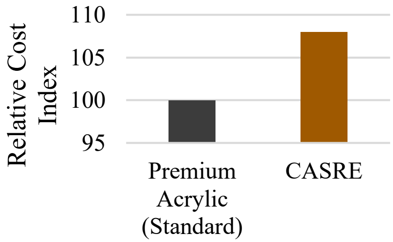
Slight binder cost increase (~5–10%)
•No additional plant CAPEX required
•Uses existing manufacturing infrastructure
•Standard pigment & filler supply chain

Pricing Strategy
•10–15% above premium acrylic segment
•Positioned as performance-enhanced coating

~25% Lower Lifecycle Cost



Value Proposition & Competitive Edge For Customers



Reduced cleaning frequency
•Lower long-term maintenance cost
•Extended repaint cycle (5–7 years)
•Fluorine-free sustainability advantage

For Manufacturer
•Incremental margin expansion
•No major R&D infrastructure shift
•Premium product differentiation